

Three Attacks on Resolution Geometry

A Stress-Test Battery, with One Concession

Matěj Rada

MatejRada@email.cz

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Abstract

Three adversarial questions remain against the resolution-geometry (RG) results. (A) Is the lock an artifact of the observation protocol—same probes, same conclusions, nothing intrinsic? (B) Does the gluing homotopy *manufacture* the atlas, so that any blending would show the same holonomy? (C) Is the holonomy real but generic—would any smooth Grassmann bundle behave the same? Each is tested numerically. The verdicts are honest and not uniform. Attack A fails: the lock is a set of algebraic identities that hold to 10^{-14} for arbitrary (J, Σ) of arbitrary dimension, and the structural conclusions (resolved rank, strict Schur floor, graded canonical correlations, non-zero saturation gap) persist across five very different probe geometries—only the *values* change, as they must. Attack B fails: holonomy is produced by curvature, not by blending—controls that move the resolved subspace by 12.8 radians but commute give holonomy ≈ 0.0015 , while non-commuting controls that move it only 7.2 radians give 2.53—and the holonomy is a reparametrization-invariant loop invariant (sampling-independent to 10^{-5}). Attack C is *partly conceded*: a generic smooth Grassmann bundle already carries a convergent, non-trivial holonomy, so RG did not invent the phenomenon. What RG adds is not conceded: a *generic* coupling is almost never even admissible as an RG geometry (0 of 400 at moderate coupling strength, against 400 of 400 for RG-built geometries), and the saturation counting law ($\dim = p + q + pq$) is RG-specific. Holonomy is generic; RG’s admissibility and stratification are not.

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1 The three attacks

Attack 1 (Protocol artifact). The lock and its conclusions are artifacts of the chosen observation protocol: *same probe geometry* $\Rightarrow$ *same conclusions*, with nothing intrinsic behind them.

Attack 2 (The homotopy manufactures the geometry). The gluing holonomy is created by the blending construction. A different embedding or a different blend would show the same atlas behavior, so the holonomy reports the homotopy, not any geometry.

Attack 3 (Not unique to RG). The holonomy is real but generic: any sufficiently smooth Grassmann bundle would show the same thing, so it is no evidence for RG specifically.

Each is answered by a battery (`Resolution_Geometry_Attacks_battery.py`, `numpy/scipy` only). We report what passed, what failed, and what we concede.

2 Attack A: is the lock a protocol artifact?

2.1 A1 — the lock is algebraic

The lock is not a numerical coincidence of one probe set; it is a chain of identities that must hold for *every* observation map J and covariance Σ . We verified them on 300 random (J, Σ) of randomly varying dimension—i.e. 300 arbitrary “protocols”:

lock identity	max error over 300 random inputs
block-inverse / Schur: $R_b^\top \Sigma^{-1} R_b = (\Sigma_R - C \Sigma_N^{-1} C^\top)^{-1}$	2.3×10^{-14}
Fisher = pullback of inverse Schur: $J^\top \Sigma^{-1} J = A^\top \text{Schur}^{-1} A$	2.7×10^{-13}
gauge invariance of canonical correlations	3.9×10^{-15}
gauge invariance of frame alignment $\text{tr}(\Sigma_R C \Sigma_N C^\top)$	3.5×10^{-10}

The identities hold to machine precision regardless of input. A relationship true for *all* (J, Σ) cannot be an artifact of one probe geometry.

2.2 A2 — the conclusions are protocol-invariant, the values are not

We then ran the actual chaotic-vortex pipeline through five deliberately dissimilar probe geometries. The distinction that matters: the RG *conclusions* are structural (a resolved rank, a positive Schur floor, a graded canonical spectrum, a non-zero orientation content), while the *numbers* legitimately depend on what the probes see.

probe geometry	p	Schur _{min}	ρ_1	ρ_4	saturation gap
3 shells $r=3, 4, 5$	6	0.074	0.78	0.30	0.85
2 shells $r=2.5, 6$	6	0.075	0.85	0.36	5.51
random scatter (24)	6	0.074	0.85	0.09	5.05
one-sided arc (20)	6	0.074	0.51	0.005	0.13
far sparse $r=8, 10$	6	0.074	0.30	0.005	0.006

Across every protocol: $p = 6$ (the rank equals the number of driving parameters, not the probe choice), the Schur floor is strictly positive, the canonical correlations are a descending grade, and the saturation gap is non-zero (frame alignment always carries information beyond ρ). The values differ—far, sparse probes resolve the coupling weakly, so their orientation content shrinks to 0.006—which is exactly correct: different observations resolve different amounts. The structure is intrinsic; the numbers are the protocol’s honest report of it.

Verdict 1. Attack A fails. The lock is algebraic (A1), and the structural conclusions are protocol-invariant while the values vary as they should (A2). “Same probes, same conclusions” is true only in the trivial sense that identities are identities.

3 Attack B: does the homotopy manufacture the geometry?

3.1 B1 — holonomy is curvature, not blending

If the homotopy manufactured the holonomy, then blending “a lot” would always produce it. It does not. We compare two two-parameter families of the resolved subspace built from skew generators Ω_1, Ω_2 : one with *commuting* controls ($\Omega_2 \parallel \Omega_1$), one with *non-commuting* controls.

controls	subspace travel around loop	holonomy
commuting	12.76 rad (large)	0.0015
non-commuting	7.15 rad (smaller)	2.53

The commuting family drags the resolved subspace nearly twice as far yet returns almost no holonomy; the non-commuting family moves it less and returns a large one. Holonomy measures the curvature (the non-commutativity) of the loop, not the amount of blending. A homotopy that “blends” without curvature manufactures nothing.

3.2 B2 — holonomy is a loop invariant

Holonomy is invariant to how the loop is drawn. Sampling the same loop uniformly versus clustered near its corners gives the same value:

samples N	uniform	clustered	diff
48	2.0092	2.0093	10^{-4}
96	2.0092	2.0093	10^{-4}
192	2.0091	2.0091	2.5×10^{-5}

The value belongs to the loop, not to its parametrization. Different *constructions* that trace different loops give different values—correct, since each is the invariant of its own loop—but the existence and geometric nature of the holonomy does not depend on the construction, as B1 already established.

Verdict 2. Attack B fails. Holonomy is produced by curvature, not blending (B1), and is a reparametrization-invariant invariant of the loop (B2). The homotopy reveals the atlas; it does not manufacture it.

4 Attack C: is the phenomenon unique to RG?

This is the attack we *partly concede*, and the concession is the honest part.

4.1 C1 — holonomy is generic (conceded)

A generic random smooth subspace bundle—Im of $M_0 + sM_1 + tM_2 + stM_3$ with random M_i —already carries a convergent, non-trivial holonomy (value 0.0956, overlap defect 0.000, stable under refinement). Holonomy is a property of smooth Grassmann bundles in general. RG did not invent it, and the gluing paper’s holonomy is not, by itself, evidence for RG over any other smooth bundle. We concede this fully.

4.2 C2 — RG’s structure is not generic (not conceded)

What distinguishes an RG geometry from “a smooth bundle” is not the holonomy but the *admissibility and stratification* the holonomy rides on.

Admissibility. An RG geometry is a sub-block of a genuine positive-definite covariance, which forces the Schur floor—equivalently, canonical correlations $\rho \leq 1$. A *generic* coupling does not satisfy this. Drawing random (Σ_R, C, Σ_N) with coupling scale α and asking how often the assembled object is admissible:

coupling scale α	0.3	0.7	1.2	2.0	RG-built
admissible (of 400)	252	26	0	0	400

By moderate coupling strength, essentially no generic coupling is realizable as an RG geometry, whereas an RG geometry (built from a joint positive-definite Σ) is admissible always. RG geometries are a thin, constrained subclass—the Schur floor is a real condition a generic bundle has no reason to meet.

Stratification. The saturation counting law is RG-specific: the fibre quotient $(\Sigma_R, C, \Sigma_N) / O(p) \times O(q)$ has dimension $p + q + pq$ on the top stratum, and the enlarged invariant set (spectral traces plus frame-alignment moments) has Jacobian rank matching it exactly—8, 15, 11 for $(p, q) = (2, 2), (3, 3), (2, 3)$. A generic bundle has no such decomposition, no walls, and no completion theorem.

Verdict 3. Attack C is partly conceded. Holonomy is generic to smooth Grassmann bundles; RG makes no exclusive claim to it. But RG’s admissibility (Schur floor $\Leftrightarrow \rho \leq 1$) and its saturation stratification are not generic—they are what RG contributes, and they survive the test.

5 What survives

The three attacks, taken together, sharpen rather than weaken the position. The lock is algebraic, not a protocol artifact. The atlas holonomy is a genuine loop invariant driven by curvature, not manufactured by the gluing construction. And the holonomy itself is conceded to be a generic feature of smooth bundles—which is why the substance of RG was never the holonomy but the *construction underneath it*: the resolved/null split of a covariance, the Schur admissibility floor, the canonical-correlation stratification, and the frame-alignment completion. Those are the constrained, RG-specific structures, and they are exactly the ones the battery finds intact. The honest reading is that RG is a particular, admissibility-constrained, stratified sub-family of smooth observation geometries—not a claim to have discovered curvature, but a claim to have identified the structure that a genuine observation covariance must carry. That claim stands.

A Battery summary

All figures from `Resolution_Geometry_Attacks.battery.py` (numpy + scipy).

- A1: lock identities, max error 2×10^{-14} – 3×10^{-10} over 300 random (J, Σ) .
- A2: five probe geometries; $p = 6$, $\text{Schur}_{\min} \approx 0.074$, graded ρ , saturation gap > 0 in all; values range widely.

- B1: commuting controls, travel 12.76 rad, holonomy 0.0015; non-commuting, travel 7.15 rad, holonomy 2.53.
- B2: same loop, uniform vs clustered sampling, $|\text{diff}| \leq 10^{-4}$, converging.
- C1: generic smooth bundle, holonomy 0.0956, defect 0.000, converged (conceded).
- C2: generic-coupling admissibility 252/26/0/0 of 400 at $\alpha = 0.3/0.7/1.2/2.0$ vs 400/400 RG-built; saturation Jacobian rank 8, 15, 11 matches quotient dimension.

References

- [1] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.
- [2] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.
- [3] A. Edelman, T. Arias, S. Smith, “The geometry of algorithms with orthogonality constraints,” *SIAM J. Matrix Anal. Appl.* 20 (1998), 303–353.
- [4] B. Simon, “Holonomy, the quantum adiabatic theorem, and Berry’s phase,” *Phys. Rev. Lett.* 51 (1983), 2167–2170.